Paleomagnetism of Archean volcanics and Proterozoic dikes in La Grande Riviere area of Nouveau-Quebec[1]

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Paleomagnetic results are reported for six Archean volcanics and six Proterozoic diabase dike sites in the La Grande Riviere basin using AF and thermal demagnetization. The area is now flooded so that future work is not possible. The limited geological and statistical data base makes interpretation of the results necessarily tentative. The dikes yield an A component pole at 84°W, 74°N from magnetization residing in higher stability domains, which likely reflects emplacement either at ~2150 or ~1825 or ~1650 Ma ago, and a B component pole at 33°W, 4°S residing in lower stability domains, which likely reflects metamorphism during the Hudsonian Orogeny at ~1725 Ma, or, less likely, during Grenvillian uplift at ~975 Ma. The volcanics appear to yield a prefolding C component pole at 28°W, 49°S that dates extrusion at ~2900 Ma and a postfolding D component pole at 306°W, 37°N that dates metamorphism by the Kenoran Orogeny at ~2600 Ma.

Les résultats d'une étude paléomagnétique sont rapportés pour six échantillons de roches volcaniques de l'Archéen et six de dykes de diabase du Protérozoïque prélevés dans le bassin de La Grande Rivière et utilisant la désaimantation par champs alternatifs et par traitements thermiques. La région est maintenant inondée, excluant ainsi les études ultérieures. Les connaissances géologiques et les données statistiques de base sont telles que l'on puisse seulement envisager une interprétation des résultats. Les dykes présentent une composante A du pôle à 84°O, 74°N d'une aimantation portée par les domaines les plus stables, suggérant leur mise en place il y a soit 2150 ou 1825 ou 1650 Ma, et une composante B du pôle à 33°O, 4°S portée par les domaines moins stables qui semblent refléter une phase de métamorphisme durant l'orogénèse de l'Hudsonien il y a 1725 Ma, ou moins possiblement durant le soulèvement grenvillien il y a 975 Ma. Les roches volcaniques fournissent une composante C du pôle à 28°O, 49°S d'une aimantation acquise antérieurement à la déformation par plissement qui indique que les épanchements datent de 2900 Ma, et une composante D du pôle à 306°O, 37°N d'une aimantation acquise postérieurement au plissement et qui date à 2600 Ma l'avènement du métamorphisme causé par l'orogénèse du Kénoranien.

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Introduction and geology

Samples for this paleomagnetic study were collected in 1978 prior to flooding of the La Grande Riviere basin (Fig. 1) by Hydro Québec power projects. The conclusions are very tentative because details on the geologic relationships between the units are not known in detail and because no radiometric age dates are available.

The La Grande Rivière area is in the Superior Structural Province (Eade 1966). The Archean metavolcanics and metasediments occur in a few east–east-northeast trending "greenstone" belts (Sharma 1974*a*,*b*, 1975, 1976, 1977, 1978). They were intruded by plutonic rocks of dominant quartz diorite to granodiorite composition. During the Kenoran Orogeny at ~2.5 Ga, the Archean strata were deformed into tight east to east-northeast trending folds that plunge at 35 ± 5°W, were metamorphosed into the mid-greenschist to amphibolite facies, and were cut by late pegmatite dikes and quartz veins. Gently dipping pink orthoquartzites of the lower Aphebian Sakami Formation unconformably overlie the Archean basement in several outliers. Quartz- and olivine-bearing gabbro diabase dikes were emplaced next to form a north-northwest trending

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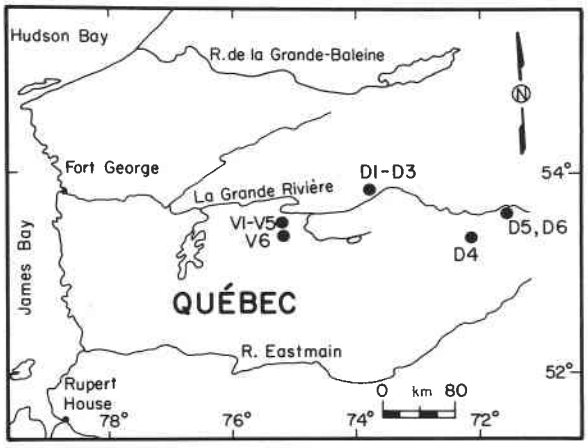


FIG. 1. Map showing the location of dike (D) and volcanic (V) sampling sites.

swarm. These dikes appear to be unmetamorphosed except for minor deuteric alteration. Extensive unconsolidated Quaternary sediments cover the area.

Sampling and measurement

The Archean metavolcanic sites include four flow sites of massive komatiitic basalt (V1, Fig. 1), komatiitic basaltic pillow breccia (V2), high-magnesium basalt (V3), and massive spinifexed komatiitic peridotite (V4), and two sill sites of serpentized peridotites (V5, V6). All six dike sites (D1–D6) are in massive diabase with ophitic texture. From two to five oriented hand or core samples were collected at each site, and from four to 20 specimens were cut from each sample. The 17 volcanic samples yielded 129 specimens and the 20 dike samples yielded 157 specimens.

The natural remanent magnetization (NRM) of each specimen was measured using a Schonstedt DSM-1 spinner magnetometer. The NRM intensities ranged between 10^2 and 10^0 A m^{-1} and the NRM directions were broadly scattered for both the volcanics and dikes. Alternating field (AF) step demagnetization was done using equipment described by Seguin (1975) in 5 mT increments up to 75 mT. The characteristic direction, isolated in several steps between 30 and 60 mT, was determined for each of 34 volcanic and 64 dike specimens. Thermal demagnetization was done using a Schonstedt TSD-1 oven at steps of 200, 300, 400, 450, 500, 540, 550, 565, 570, and 600°C. The characteristic direction isolated in the 540–565°C range was determined for each of 95 volcanic and 93 dike specimens.

Analysis of directions

The characteristic directions for the dikes and volcanics were plotted on equal-area stereonets and their point density was contoured using a 1% smoothing area (Kamb 1959; Van Alstine 1980; Stupavsky and Symons

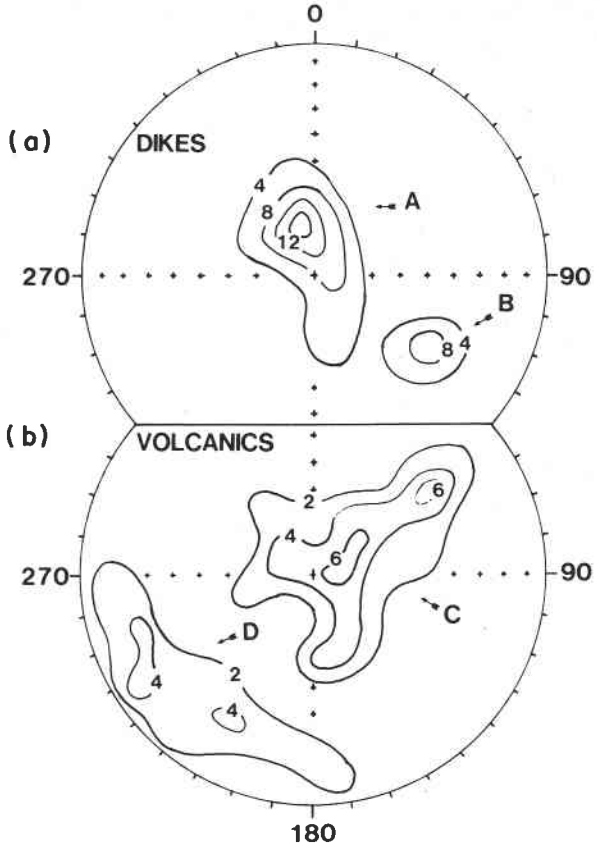


FIG. 2. Stereocontour plot on a Schmidt equal-area net of the combined AF and thermal demagnetized end-point directions for (*a*) diabase dikes after rotation of negative vectors by 180° with the contours in 4% point density increments, and (*b*) volcanics after rotation of positive vectors by 180° with the contours in 2% point density increments. The peak values for components A, B, C, and D are 19, 10, 7, and 5%, respectively.

1982*b*) using AF data only, thermal data only, and AF plus thermal data. Both the dikes and volcanics were found to have two positive nodes and two antiparallel negative nodes in both the AF and thermal data. The dikes have a dominance of positive inclinations (53%) so their negative directions were rotated to their antiparallel positive position. Conversely, the volcanics have a dominance of negative inclinations (56%) so their positive directions were rotated to their antiparallel negative positions. Every site had at least one sample of each polarity (positive and negative), and the reasons for this are not clear. The resulting stereocontour plots are shown in Fig. 2.

For the dikes, all directions falling within the 4% density contour bounds were used to get the A and B component directions and, for the volcanics, the 2% bounds were used to get the C and D component directions. The first approach was to combine the

specimen directions to obtain the sample/core means, which were combined to obtain the site means, which were combined to obtain the unit means (Table 1). The second approach was simply to combine the specimen means to determine the unit means. A number of points should be made. Except for site D1 with A component directions only, all of the dike sites retain both A and B components of magnetization. Similarly, all of the volcanics sites retain both C and D components. In both data sets, many samples retain both components. The AF and thermal data sets give statistically similar mean directions for the A and B components in the dikes and for the C and D components in the volcanics, with the thermal data giving substantially less dispersion in both units. The pole positions for the four components have been calculated conservatively using the site mean directions from the combined AF and thermal data (Table 2).

TABLE 1. Summary of remanence directions

Unit, component, average method	AF data						Thermal data						Combined AF and thermal data					
	N	n	n^1	$D°$	$I°$	$\alpha_{95}°$	N	n	n^1	$D°$	$I°$	$\alpha_{95}°$	N	n	n^1	$D°$	$I°$	$\alpha_{95}°$
Dikes: outliers	2	6	9				5	6	13				5	9	22			
A, sites	5	12	39	358	74	20	6	15	59	334	83	11	6	19	98	352	79	11
A, specimens			39	4	66	7			59	340	80	6			98	350	75	5
B, sites	3	3	16	142	53	23	3	6	21	131	34	20	5	9	37	136	41	15
B, specimens			16	127	37	6			21	130	35	9			37	129	36	6
Volcanics: outliers	5	10	19				6	13	28				6	13	47			
C, sites	6	8	9	70	−69	37	6	14	40	75	−65	17	6	18	49	73	−65	19
C, specimens			9	80	−64	27			40	76	−71	9			49	77	−70	9
D, sites	3	4	6	211	−33	54	5	12	27	226	−19	16	6	13	33	220	−21	20
D, specimens			6	223	−33	30			27	220	−23	11			33	221	−25	10

NOTES: N, n, and n^1 are the number of sites, samples/cores, and specimens, respectively. D, I, and α_{95} are the declination, inclination, and radius of the cone of 95% confidence (Fisher 1953), respectively.

Interpretation

Dikes

Both of the A and B dike components are removed by thermal cleaning at 600°C, indicating that both probably reside in magnetite. Both also survive AF cleaning at 30–60 mT equally well; however, the B component is preferentially removed by 555 ± 15°C thermal cleaning. Polished section examination reveals that both coarse- and fine-grained magnetite populations are present in the diabase. Thus it appears that the more stable A component resides preferentially in the higher coercivity fine-grained magnetite, and the less stable B component in the lower coercivity coarse-grained magnetite. This implies but does not prove that the A component is older than the B component. Given that the dikes show no evidence of metamorphism and that the flat-lying Sakami Formation that they intrude shows a maximum burial metamorphism of sub-prehnite–pumpellyite facies temperatures (<250°C), the A component is considered to be primary or acquired on emplacement, and the B component is considered to be secondary and acquired on orogenic uplift after burial.

The age of the Sakami Formation is uncertain. Geologically, the unit is similar to pink argillites and quartzites in the Gowganda and Lorrain Formations of the Huronian, which are dated at ~2250 Ma (Roscoe 1973). Seguin *et al.* (1981) reported a postfolding Sakami pole of 103°W, 18°S, which they correlated with equatorial Nipissing diabase, Gowganda sediment, and Abitibi diabase poles dated at ~2200 ± 50 Ma (Symons *et al.* 1981). If true, then the A component dike pole of 84°W, 74°N likely correlates with the northern group of Nipissing, Gowganda, and Abitibi poles dated at ~2100 ± 50 Ma. Alternatively, the Sakami Formation could correlate with the Richmond Gulf red beds or the Seward–Sokoman red orthoquartzites of the Circum-

TABLE 2. Pole positions of remanence components

Unit, component	Pole position			
	Long. (°W)	Lat. (°N)	Oval of 95% confidence δp°	δm°
Dike				
A	84	74	20	21
B	33	−4	11	18
Volcanics				
C (uncorrected)	121	28	25	31
C (corrected)	28	−49	15	28
D (uncorrected)	306	37	11	21
D (corrected)	273	20	12	22

NOTE: δp and δm are the radii of the oval of 95% confidence.

Ungava geosyncline. The Seward A pole of 94°W, 23°S was considered by Park (1977) to be 1825 ± 75 Ma old. Schwarz and Freda (1980) reported a prefolding Sakami pole of 108°W, 10°N, which they correlated with a number of poles including the Seward B pole that Park (1977) and Irving (1979) considered to be post-Hudsonian and placed at ~1650 ± 75 Ma. If the Sakami is either ~1825 or ~1650 Ma old, then the A pole for the dikes was likely set on emplacement at ~1600 ± 50 Ma based on present estimates of the apparent polar wander (APW) path (Irving 1979; Stupavsky and Symons 1982*a*).

Assuming that the Sakami pole is pre-Hudsonian at ~2200 or ~1825 Ma, then the dike B component pole of 33°W, 4°S fits neatly on the Hudsonian loop of the APW path at ~1725 Ma. This fits with the high blocking temperatures of up to 550°C for the component. Conversely, the B antipole also lies near the Grenville Orogeny loop of the APW path at ~975 Ma and could be an uplift component. This fits with the subgreenschist alteration at most of the dikes and Park's (1977) observation of similar Grenvillian uplift components in most rock units of the Labrador Trough.

Volcanics

The C and D components of the Archean volcanics are both thermally unblocked at ~555 ± 15°C. Although the C component is more prevalent than the D, both appear equally stable, and hence either may be older. Also, their high blocking temperatures suggest that both were set before or during the Kenoran Orogeny, which deformed and regionally metamorphosed the volcanics prior to ~2450 Ma. The volcanics strike at 260 ± 10° and dip at 62 ± 8°S at all sampling sites, so that the fold test is equivocal. Microfold structures show that the volcanics plunge at 35 ± 5°W. The pole positions for the C and D components are given in Table 2 both before and after tectonic correction for attitude and plunge.

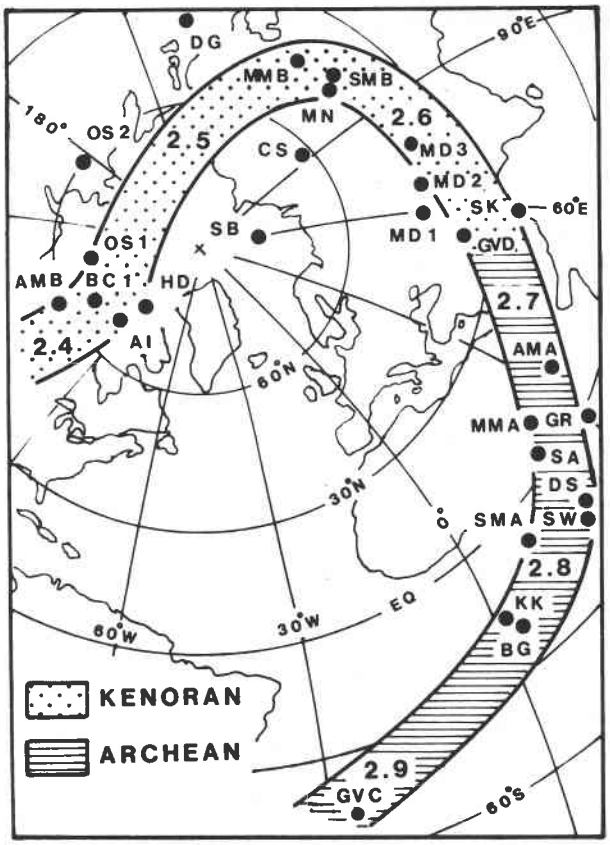


FIG. 3. Apparent polar wander path for North America between 2.9 and 2.4 Ga. The poles are identified and summarized by Irving (1979), Strangway *et al.* (1981), and Symons *et al.* (1981). GVC and GVD are the Grande Riviere volcanic poles determined in this study.

The uncorrected C component pole does not fall on any part of the APW path that is indicative of a Kenoran, Hudsonian, or Grenvillian metamorphic overprint. Conversely, the corrected C pole of 28°W, 49°S falls directly on the logical extension of the APW path to ~2900 Ma (Fig. 3), and may therefore record primary extrusion of the volcanics. The decreased scatter of the C directions on tectonic correction favours this interpretation. If true, then this is one of the oldest poles yet determined for the Canadian Shield. The uncorrected D component pole of 306°W, 37°N falls on the APW path at ~2600 ± 50 Ma or early in the Kenoran Orogeny segment, but falls well away from any younger orogenic segment. The corrected D pole falls well away from any portion of the pre-Kenoran APW path unless by unlikely chance it is very much older than the C component. Therefore, the D component is most likely a postfolding Kenoran metamorphic remagnetization acquired ~2600 Ma ago.

Finally, plots showing the demagnetization behaviour of example specimens and detailed tables providing the

site remanence statistics are available from one of us (D.T.A.S.) on request.

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